POTPRESS · RISK REGISTER

PotPress risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	P2 and H10 at the 5.4 t working force limit decided by Amish (D9, 2026-10-25)
0.2	2026-10-23	Amish Chadha	Hazards and risks updated for the STEM check fixes (PPR-DDR-006); eye bolt hazard H11 added
0.1	2026-10-23	Amish Chadha	First risk register, hazard analysis and design FMEA (kit 1.8 upgrade), including the four risks the independent STEM review found

PotPress risk register and hazard analysis

The top project risk is that the forming force and clay shrinkage are unknown until pressing trials with a partner, which could change the mold size and the operator effort. The top hazard is crushing at the molds, controlled by the fixed guards and the two interlocks; the highest design FMEA scores are charge mass against the stop and side load on the jack. Method: STANDARDS section 23 and ISO 12100. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (unlikely to be caught).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No partner factory	Partners are picked per area later (PRB open questions)	Requirements stay unvalidated; no TRL 4	3	5	Amish	Keep the design at TRL 3; approach a network factory when TRL 4 opens
P2	Forming force higher than expected	Not published; one press forms at 2 t	Working force capped at 5.4 t by the handle (D9, decided 2026-10-25); cycle time rises	3	3	Design lead	Trials start near 2 t; separate two-speed pump (D6, proposed, still open) if above 5.4 t
P3	Fired pot size wrong	No clay shrinkage allowance (STEM M6)	Pots do not fit receptacles	2	4	Design lead	Molds scaled for 8 % assumed shrinkage (PPR-DDR-006); test bar on the partner clay before patterns are printed
P4	No foundry within reach	Aluminium casting is city-based	Molds must be imported or made another way	3	4	Partner	Ask at partner selection; keep pattern files open for any foundry
P5	Cost stays over target	Guarding and construction details	Weaker case against imported presses	4	2	Amish	Value-engineering list in PPR-DEC-001
P6	Guard check not signed	Needs a competent person	No force above hand pressure	2	4	Amish	Find a signatory with the partner before TRL 4

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Crushing between the molds	Hand in the press while pumping	Crushed hand or fingers	5	2	Fixed guards to ISO 13857, gate interlock on the release, controls outside the guard	1 (desk check unsigned)
H2	Shear at the platen guides and carriage	Reaching through the gate gap	Finger injury	4	2	6.35 mm mesh on front, gate edge gaps checked	1
H3	Load pin not home	Pin left out after a mold change	Male mold slips under load	5	2	Pin-presence interlock	1
H4	Ejected part under load	Bolt or pin failure at 20 t	Impact injury	5	1	1.5 times rating without yield; roof guard	1
H5	Hydraulic oil release	Seal failure or overpressure	Skin injection, slip, clay contamination	3	2	Jack's own relief valve; drip tray under the jack (proposed)	2
H6	Manual handling of heavy parts	39.6 kg platen and female mold, 38 kg top beam	Back injury	3	3	Chain block for every part over 25 kg; eye bolts in the female base plate and male flange; holes plugged before pressing (PPR-DDR-006)	1
H11	Eye bolt left in a mold	Mold change not finished	Bolt crushed under the male flange or striking the plug as the carriage slides; ejected fragment	4	2	Eye bolts out and holes plugged before the press moves (build plan S4); female lift points in the base plate, not the stop face	1
H7	Press tipping	Pushed or pulled while unanchored	Crushing	5	1	Four M12 floor anchors; tipping moment 844 N m checked	1
H8	Silica and burnout dust	Dry clay and rice husk ash	Lung disease over years	4	3	Wet methods, masks; the factory's process (STEM B3)	2
H9	Zinc fume	Welding galvanized mesh frames	Metal fume fever	3	2	Grind off zinc at welds, ventilation (STEM B4)	1
H10	Repetitive strain	Pump handle force at high settings	Shoulder and wrist strain	3	3	Working force limited to 5.4 t (about 148 N with 85 % pump efficiency; D9, decided by Amish 2026-10-25); no handle extensions; EN 1005-3	2

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						assessment to do (PPR-DDR-006)	

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Clay charge	Fill the cavity	Charge too heavy for the stop	Molds do not reach the stop; wall too thick; frame sees full jack force	Charge weighed by eye; mix moisture varies	4	4	3	48	Weigh each charge to a tolerance found in trials; add charge tolerance to PR-002
F2	Bottle jack	Make the force	Side load on the ram	Seal wear, leak, bent ram	Platen guides bind; female mold off centre	4	3	3	36	Check guide clearance against jack side-load limit; centre the adapter disc
F3	Floor anchors	Hold the press	Anchor pulls out	Press tips when the carriage is pulled	Weak or cracked floor	5	2	3	30	State minimum floor (concrete grade and thickness) in the build plan; pull test at install
F4	QC rack	Measure flow	Falling-head reading taken as steady flow	Wrong accept or reject	Flow falls as the head drops over the hour	3	3	3	27	State the head and time the band refers to; same fill height every test
F5	Molds	Set the wall	Cavity off size after casting	Uneven wall; PR-002 not met	Shrink, porosity, template error	4	3	2	24	Template check after finishing; print error in the tolerance budget (STEM E2)
F6	Joint bolts	Hold the frame	Loosening under cycles	Frame misaligns	63 MPa stress range near the 71 MPa fatigue limit (STEM S9)	4	2	3	24	Hardened washers, torque marks, retorque at intervals
F7	Gate interlock	Stop pressing with the gate open	Interlock defeated	Hand reaches the molds under force	Operator ties it open	5	2	2	20	Assess against ISO 14119 defeat rules (G3)
F8	Return springs	Lower the platen	Spring breaks	Platen stays up; demolding slows	Fatigue, corrosion	2	2	1	4	Spares in the kit